

Time-Series Modeling of Mental Health Indicators in the United States During and After the COVID-19 Pandemic

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Introduction

The Covid-19 pandemic has had a significant and far-reaching impacts on global society, with outcomes that are still propagating even in 2024. One particular example of this, and an area that had a particularly significant impact, is the topic of mental health.

Mental health has always been an important topic to consider, but has not been taken as seriously as it should be considering how its impact. This problem of a lack of acknowledgement surrounding mental health was likely due to the problem not having been as widely discussed prior to 2020. In fact, a household survey done in 2019 by the National Center for Health Statistics found that only between 8-10 percent of the those surveyed believed they had either anxiety, depression or some degree of both. Considering this, it is possible that the topic was simply not seen as that big of a concern, even though it is difficult to say how accurate those numbers even were. As we will see shortly into the paper, these numbers will then also begin to heavily skyrocket once the pandemic begins.

The aim of our project then is to widen the awareness of the topic of mental health and how far reaching the health issue is across the United States, particularly due to the Covid-19 pandemic throughout the past four years. To do this, we used a dataset initiated by the CDC which surveyed a number of households across the country in partnership with the Census Bureau to understand how households were feeling due to the regulations and restrictions imposed during the pandemic. While the logistics and results of the dataset will be discussed next, it should be noted the percentage of those who believed they had some form of anxiety or depression across the country

spiked from 8-10 to upwards of 40 percent just in May of 2020. This therefore is already showing just how much of an impact the pandemic has had on mental health

What is the Dataset?

As mentioned, the dataset we are using is a survey on indicators of mental health across households over the past four years, beginning at the end of April in 2020 and continuing into 2024 as recently as April. Using the census bureau, the CDC questioned various households across the country of differing ages, gender identities and race to check whether they felt like they had any symptoms of anxiety, depression or some degree of both. This was done through a range of questions, unfortunately we have not been able to find the exact questions asked. The survey was initially sent out every week for around 6 months, and then transitioned to every two weeks for around two years before being sent out every two weeks but with a two week gap in between. This means there is a significant amount of data that we have been able to use to study the topic and how it has shifted over the past four years.

Before implementing any predictive models on the dataset, we had to consider whether any data cleaning was necessary to avoid any potential errors or complications that would lead to inaccurate results. Most of the dataset did not require cleaning, except for several small points of missing data (a period of a week or two) and one large missing segment from November 2023 till January 2024. The small data was filled using the data before and after the period, while the larger segment was ignored as there was no reliable way we found of filling in the segment accurately.

Summary Statistics

The analysis revealed that the data shows larger variation in mental health indicators, specifically anxiety and depression, during the years 2020-2023 compared to 2024. Over these years, there was a noticeable decrease in the mean and median values of anxiety and depression, suggesting an overall improvement in mental health as the pandemic progressed.

Demographic analysis provided further insights. Variations in mental health indicators were evident based on different identity factors such as gender, age, and socioeconomic status. Racial differences also played a significant role, with anxiety and depression levels varying among different racial groups. Geographic variations were also notable, with certain regions displaying higher or lower mental health issues.

Exploratory Analysis

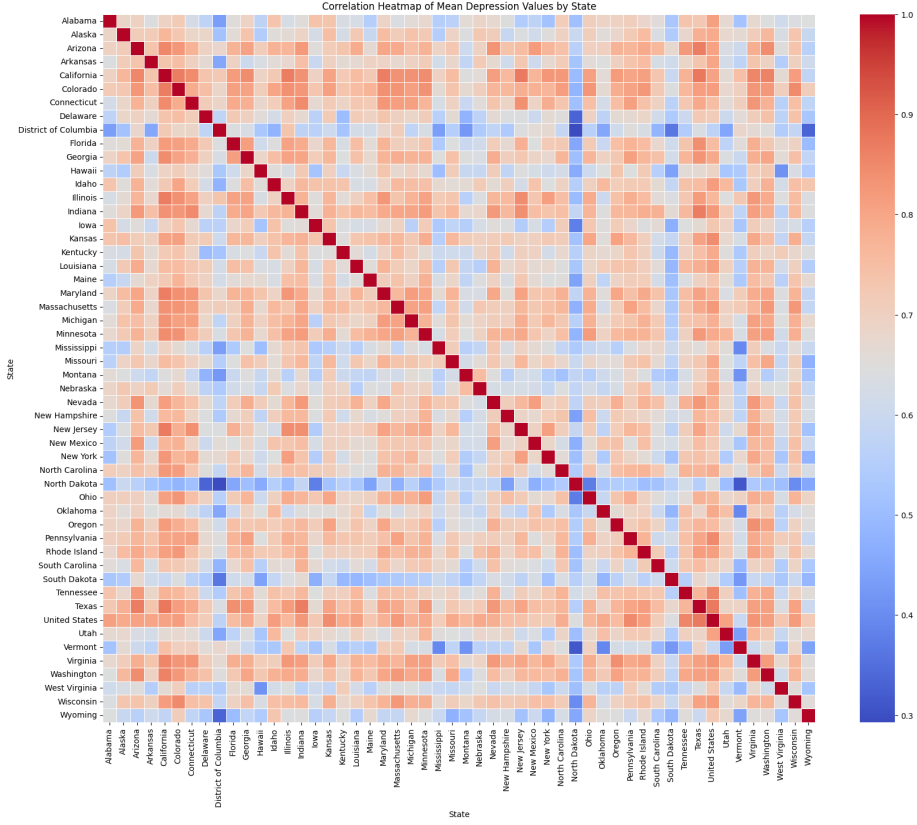


Figure 1: Heatmap comparing the mean data values between states. The means are represented as a range 0 to 1 as shown by the color bar. States compared to themselves are indicated by the diagonal ($\bar{X} = 1$).

Our exploratory analysis began with an initial examination of the dataset to identify key trends and patterns. This step was very important when it came to guiding the direction for more in-depth analysis. By understanding the general trends of the data, we were able to pinpoint specific areas that required further investigation.

Our heatmap analysis (Figure 1) provides insights into the correlations between different regions. For instance, we found high correlations between California and New York, likely due to similar socio-economic factors and levels of urbanization. Florida and Alaska, on the other hand, showed high negative correlations, possibly influenced by their differing climates and population densities. Interestingly, there

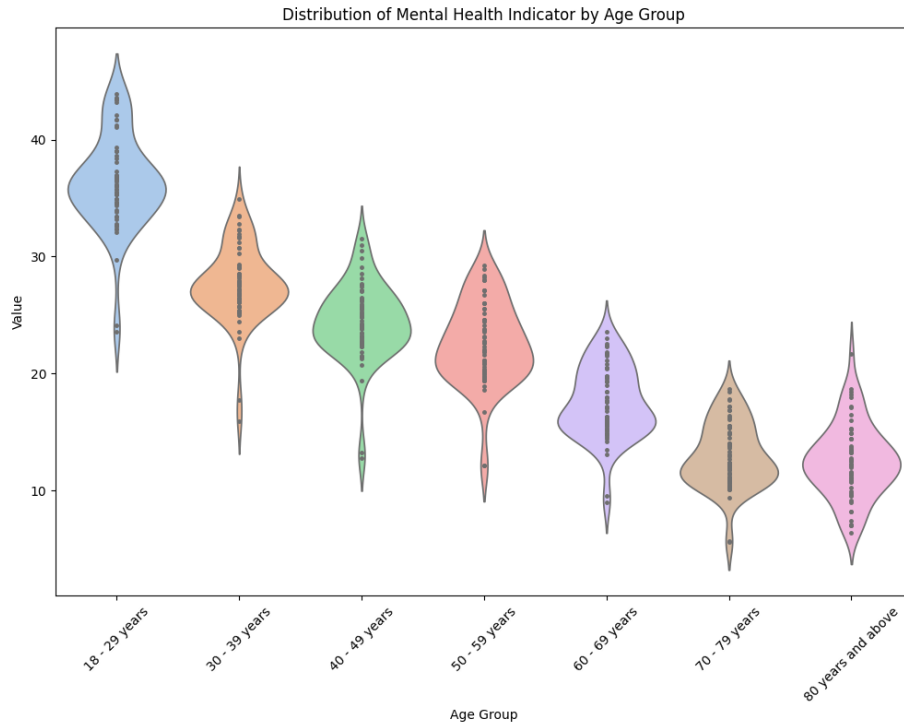


Figure 2: Violin plot describing the distribution of mental health statistics by age group. Where age is described with ten year intervals and mean mental health symptoms decrease with age

was no significant correlation between states like Texas and Vermont, suggesting that demographic and public health policy differences play a significant role.

Next, our violin plot analysis reveals the distribution of mental health indicators across different age groups. We observe that the age group of 18-29 years exhibits the widest distribution, indicating greater variability in mental health status. On the other hand, middle-aged and older adults display more consistent mental health indicators, with values clustering around the mid-30s. This suggests that younger adults experience more fluctuations in their mental health compared to older age groups. To further visualize our analysis, we developed an interactive map to observe mental health data across different regions. This tool allows users to dynamically explore mental health trends and disparities on a regional level. By providing a visual representation of the data, the interactive map can help identify areas with significant mental health challenges and guide targeted interventions

to address these issues effectively.

Predictive Models

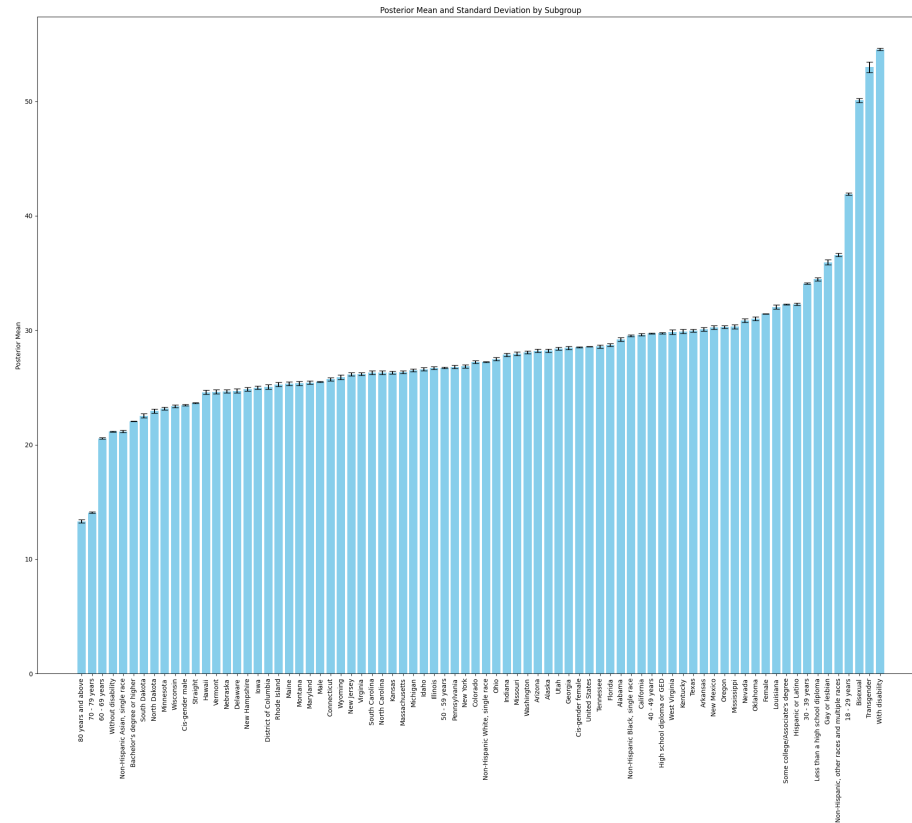


Figure 3: Bayesian model describing posterior means and variances by state or group. Black bars represent the standard deviation from the predicted mean.

For our predictive analysis, we initially utilized a Bayesian model. This model (Figure 3) updates its predictions with each new data point to produce a refined mean for subsequent data points, providing relative confidence intervals and offering insights into the uncertainty of predictions. The model performed reasonably well in tracking the overall trend of mental health indicators, showing high values for people who identify as transgender, racially mixed citizens, as well as young adults falling within the ages 18-29, and lower values for citizens 60 and above, people without disabilities, and people who have obtained a bachelors or higher. However, the root mean square

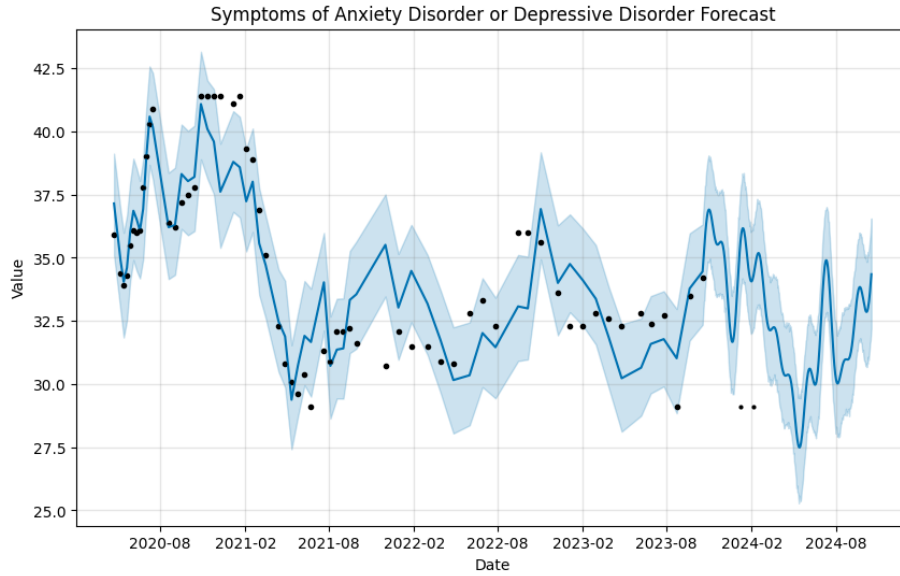


Figure 4: Prophet model predicts the symptoms of anxiety or depressive disorder up until the year 2025. Blue lines are predicted values. Blue-gray highlighted section around the line is the confidence interval. Black points are actual values.

error (RMSE) of approximately 6.35 suggested that there was room for improvement. The plot illustrates how the model's confidence intervals capture the variability in the data, yet the predictions could benefit from further refinement. Our initial Bayesian model was used to update the mean prediction with each new data point, showing a relative confidence interval with an RMSE of approximately 6.35. This model, while insightful, was less accurate than desired.

The Prophet model (see Figure 4), designed for time-series data, was employed due to its robustness with seasonal data and handling of missing values. The model was trained on data from 2020-2022 and tested on 2023 data, achieving an RMSE of approximately 3.82. The training data spanned from 2020 to 2022, and the test data was from 2023, utilizing tools such as Pandas, Prophet, Matplotlib, and Sklearn. The Prophet model effectively captured seasonal trends, offering a closer approximation of actual data trends compared to the Bayesian model.

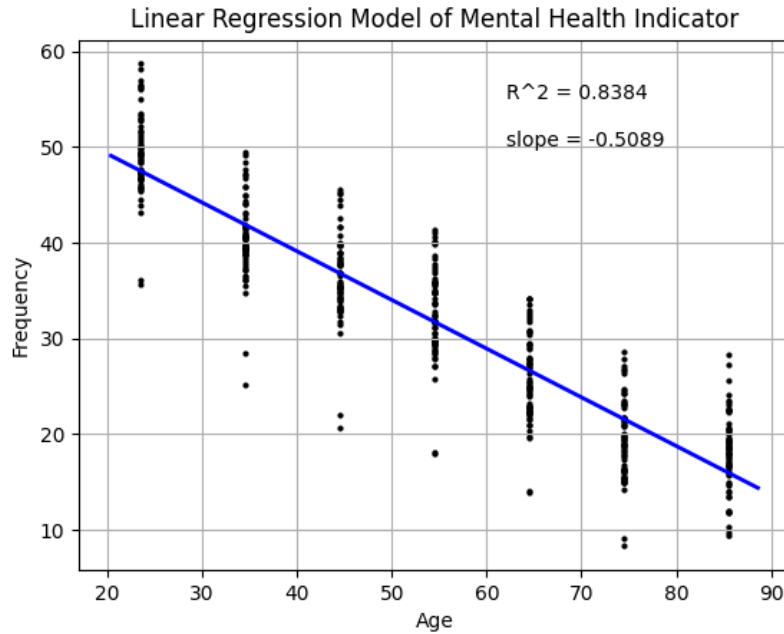


Figure 5: Least Squares Regression Model which plots the percentage of people who exhibit symptoms of Anxiety Disorder or Depressive Disorder.

The Linear Regression model shows that for each year older someone is, they test around -0.5089 lower for symptoms of Anxiety Disorder or Depressive Disorder on average. The R-Squared value of 0.8384 means that around 84 percent of the variation in the data can be explained by the linear model. The root mean square error is 20.8308. This is a relatively high value and shows that the model's predictions are less accurate.

Conclusions and Future Work

This study highlights the importance of robust data analysis and predictive modeling in understanding and managing the mental health impacts of global crises. The interactive map and seasonal trend analysis provide valuable insights, but there is a need for more data to improve accuracy. Future work should explore the impact of additional factors, such as temperature trends, on mental health outcomes.

Citations and References

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